

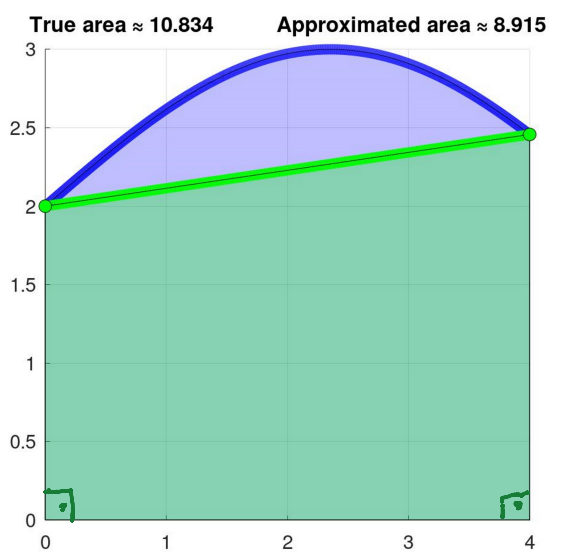
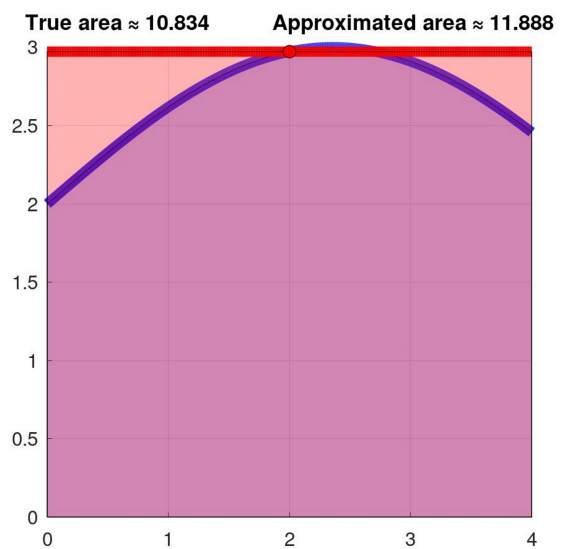
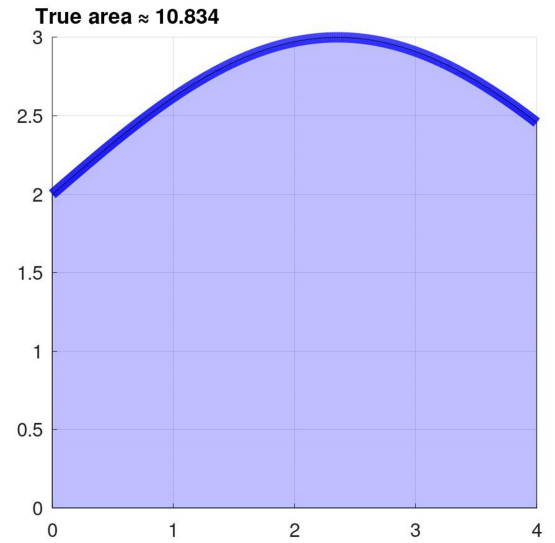
$$\int_a^b f(x)dx \approx ?$$

Formula dreptunghiului:

$$\int_a^b f(x)dx \approx f\left(\frac{a+b}{2}\right)(b-a)$$

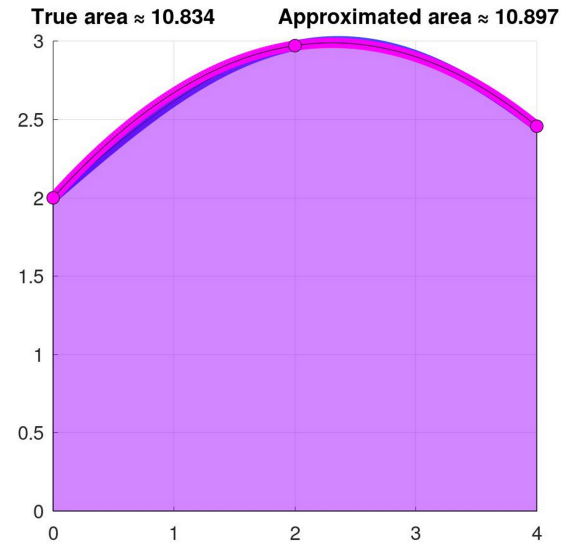
Formula trapezului:

$$\int_a^b f(x)dx \approx \frac{f(b) + f(a)}{2}(b-a)$$



Formula lui Simpson:

$$\int_a^b f(x)dx \approx \frac{b-a}{6} \cdot \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right)$$



Schema lui Romberg

$T_{1,1} =$ form. rep. a trapez. pe 2^0 subinterv.

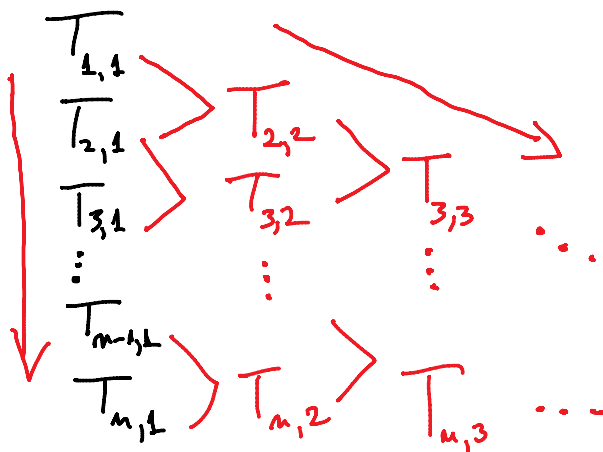
$T_{2,1} =$ ——— $||$ ——— 2^1 ——— $||$ ———

$T_{3,1} =$ ——— $||$ ——— 2^2 ——— $||$ ———

$\vdots$

$\vdots$

$T_{m,1} =$ ——— $||$ ——— 2^{m-1} ——— $||$ ———



$$T_{i,j} = \frac{4^{j+1} T_{i-1,j+1} - T_{i,j+1}}{4^{j+1} - 1}$$

$$i = 2, \dots, m$$

$$j = 2, \dots, i$$

$T_{m,m}$